

Module Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste				Paste Mask	
	F.Silkscreen			Direct Printing	Legend	GBR
	F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Ref)	0.07mm (2.00oz)		Plane	GBR
	Prepreg		0.1mm	FR4_7628	Dielectric	
	Copper	Virtual (Top side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
	Core		1.21mm	FR4	Dielectric	
	Copper	Virtual (Bot side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
	Prepreg		0.1mm	FR4	Dielectric	
	Copper	L2 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen			Direct Printing	Legend	GBR
	B.Paste				Paste Mask	

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)
Note: external layer thicknesses are specified after plating

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

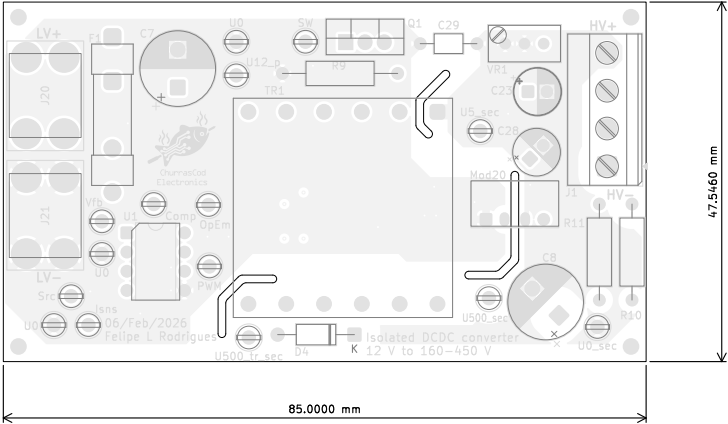
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR TOP: #8071501A/ BOTTOM: GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY FR.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	85.000 × 47.546 mm
BOARD THICKNESS	1.660 mm
TRACE WIDTH	0.400 mm
TRACE TO TRACE	-0.000 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	2.100 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.200 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

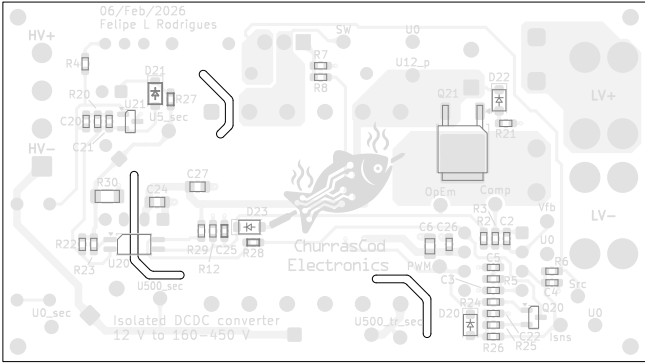


All dimensions are in millimeters unless otherwise specified.

Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR		Variant: PRELIMINARY		Git Hash: f30bd39	
	Board Name: Module		Project Name: Isolated step-up 12:450 V			
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-02	Revision: 1.0.0+ (Unreleased)	
	Sheet Path:		Reviewer: NA	Size: A4	Sheet: 1 of 7	

Module Fabrication Document

Bottom Fabrication (Scale 1



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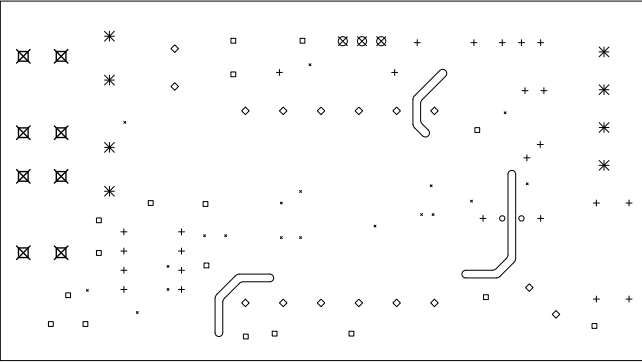
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	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Bottom Fabrication (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-02	1.0.0+ (Unreleased)
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Module Fabrication Document

Drill Drawing L1 - L4 (Scale

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	19	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
+	25	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
□	17	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	16	1,00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊠	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
✱	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊠	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
Total 98						



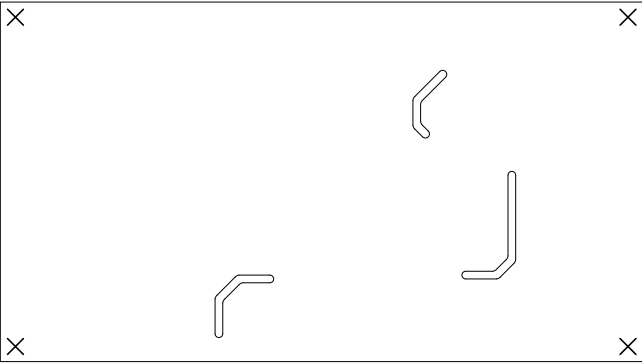
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Module Fabrication Document

Drill Drawing L1 - L4 (Scale

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.10mm (82.68milis)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 4					

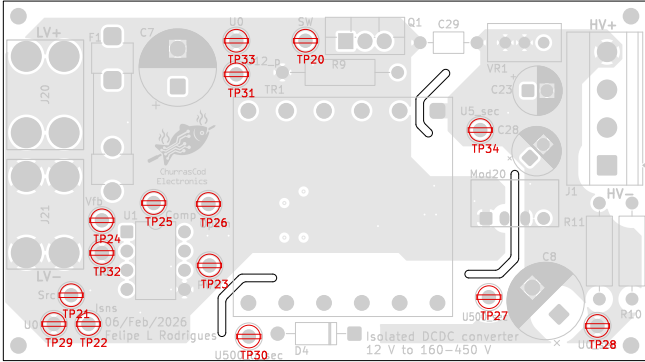


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Module Fabrication Document

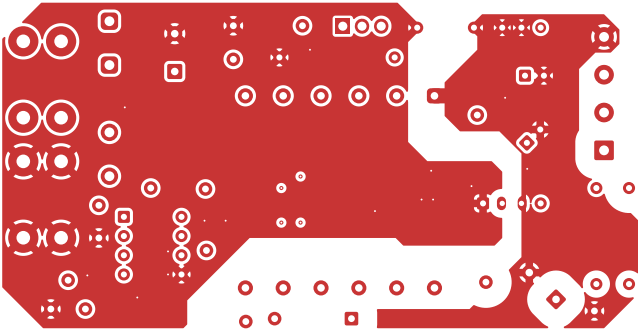
Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP20	SW	40.95	43.30
TP21	Src	9.97	9.65
TP22	Isns	12.25	5.84
TP23	PWM	28.25	13.58
TP24	Vfb	14.03	19.55
TP25	Comp	20.89	21.84
TP26	OpEm	28.13	21.71
TP27	U500_sec	65.21	9.39
TP28	U0_sec	79.56	5.58
TP29	U0	7.68	5.84
TP30	U500_tr_sec	33.46	4.18
TP31	U12_p	31.81	38.86
TP32	U0	14.03	15.23
TP33	U0	31.81	43.30
TP34	U5_sec	64.07	31.49

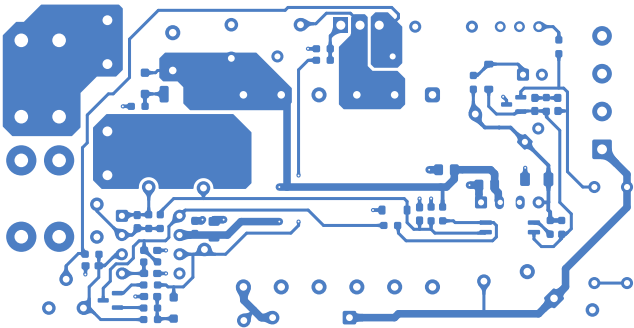


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